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Smart Soldering Workbench & Safety Guardian

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***Abstract*—Soldering work exposes hobbyists and engineers to hazards that are easy to overlook: prolonged inhalation of rosin-core flux fumes, accidental contact with hot iron tip and disasters caused by soldering irons that have been left unattended. Our project presents the design and implementation of a Smart Soldering Workbench & Safety Guardian, a benchtop safety-monitoring system built around the STM32F103C8T6 (Blue Pill) microcontroller. The system continuously fuses readings from four sensors — an MQ-135 gas sensor sampled through the on-chip ADC, an FC-51 infrared proximity switch read as a digital GPIO input, an MPU-6050 accelerometer polled over I2C, and a MAX6675 K-type thermocouple module read over SPI, to judge whether the iron is docked, in motion, dangerously hot, or surrounded by rising fume levels. Depending on this fused state, the microcontroller updates a 0.96" I2C OLED status display, switches an L293D-driven exhaust fan, drives an SG90 servo that opens or closes a vent flap through timed PWM pulses, and sounds an active buzzer if the iron is left idle off its dock for too long. Additionally, hardware telemetry and status updates are transmitted from the STM32 to an ESP8266 IoT gateway via UART protocol. The gateway broadcasts real-time telemetry at 1-second intervals via MQTT (HiveMQ) to an interactive web dashboard, while simultaneously logging periodic health and usage data at 10-minute intervals to Google Sheets using Google Apps Script Web Apps. The resulting architecture effectively converges localized hazard mitigation and remote observability, offering a robust blueprint for next-generation smart laboratory environments.**

***Index Terms*—Embedded Systems, IoT, ARM Microprocessor, Communication Protocols**

I. Introduction

Soldering is a very common process in electronics laboratory, prototyping and circuit making. Even though it is very useful, there are also some safety issues while using soldering iron. A soldering iron works at a very high temperature and even after using it, it can remain hot for some time. Also during soldering, the flux can produce fumes which may affect the air quality of the working place. Sometimes if the soldering iron is left unattended, it can also cause burn, damage of equipment or even fire. Usually in normal soldering stations, the user has to notice all these conditions by themselves and take action. So, making a smart system which can monitor these conditions and automatically take action can make the soldering workplace more safe.

A. System Workflow

The Smart Soldering Workbench & Fume Extraction Guardian is designed to make the soldering environment safer by using different sensors, actuators and communication systems together. The main controller of this system is STM32F103C8T6 microcontroller. It takes the information from different sensors and makes decisions according to the condition. In this project, MQ-135 gas sensor is used for detecting the soldering fumes and changes in air quality. FC-51 infrared sensor is used to detect whether the soldering iron is docked in its holder or it is removed. MPU-6050 accelerometer is used to detect the movement of the soldering iron and MAX6675 thermocouple module is used for measuring the temperature. These components communicate with STM32 using ADC, GPIO, I2C and SPI communication.

TABLE I
Components, their Protocols & Functions

Sensor/Actuator	Protocol	Function
FC-51	GPIO	Check for Docking Status
MQ-135	ADC	Check for Fumes
MAX6675	SPI	Check Temperature
MPU-6050	I2C	Check for Movement
OLED Display	I2C	Show Status
SG90 Servo Motor	PWM	Ventilation Open/Close
5V DC Fan	GPIO	Exhaust Fan
Active Buzzer	GPIO	Alarm

We also developed a website for monitoring the system from remotely. The website shows the live data of the soldering station such as temperature, fume condition, docking status

and movement condition. A Wemos D1 Mini is used for Wi-Fi communication, where it receives the data from STM32 through UART and sends the data to the MQTT broker. Then the website receives these data through MQTT and shows the current condition of the soldering station in real time. The website also has a live telemetry log where the received data can be seen. If the soldering iron stays undocked and does not move for a long time, the website also gives a warning to alert the user. We also store some of the sensor data in Google Sheets so that the previous data can be checked later. By using the website, the user can monitor the soldering station from another place and can easily understand the current condition of the system.

B. Operational Sequence

When the system starts, first the MQ-135 gas sensor gets some time for warm up. After that, it takes multiple readings to find a clean air baseline. During the system operation, the gas sensor continuously takes readings and compares them with a threshold value. The threshold can also change according to the normal environment, so the system does not only depend on one fixed value. When soldering fumes are detected, the system turns on a DC exhaust fan by using L293D motor driver so that the fumes can be removed from the working area. An SG90 servo motor is also controlled by PWM and it is used for controlling the ventilation system depending on the soldering iron condition.

For making the safety system more effective, the project also checks the docking, movement and temperature of the soldering iron. The FC-51 sensor checks if the soldering iron is in the holder or not and the MPU-6050 checks if the iron is moving. If the soldering iron stays outside from the holder without any movement for a certain amount of time, the active buzzer gives an alert to the user. This helps to warn the user if the soldering iron is accidentally left unattended. When the iron is placed in the holder, the MAX6675 checks the temperature to understand whether the soldering iron is still hot or it has become safe. The SSD1306 OLED display shows different information such as docking status, movement condition, fume status and temperature warning. So the user can easily understand the current condition of the system.

C. IoT & Data Logging

The project also includes UART communication and IoT system for remote monitoring. STM32 collects important data such as temperature, fume condition, docking state and movement state and sends them through UART communication. An ESP8266/Wemos module receives these data and connects the system with Wi-Fi. The data can be sent by MQTT and shown on a web-based dashboard for real-time monitoring. Some of the data can also be stored in Google Sheets, so the user can keep a record of the soldering workbench condition for later use. In this way, the system not only gives local safety protection but also provides remote monitoring and data storage.

Overall, the Smart Soldering Workbench & Fume Extraction Guardian is a low-cost and smart safety system for soldering workbench. It combines different features such as fume detection, temperature monitoring, docking detection, motion sensing, automatic ventilation, inactivity warning, OLED display, wireless communication and data logging in one system. Through this project, different embedded system concepts such as ADC, GPIO, I2C, SPI, PWM, UART, sensor control, actuator control and IoT communication are used together. The main purpose of this project is to make the soldering environment more safe, smart and easy to monitor for electronics laboratory and prototyping works.

II. Literature Review

A. State-of-the-Art

Soldering with rosin-based flux can create different health risks because fumes are produced when the flux is heated. These fumes can contain different chemical compounds and very small particles such as PM2.5, which can affect the respiratory system if a person is exposed to them for a long time [1], [2]. The UK Health and Safety Executive also considers rosin-based soldering fumes as harmful and recommends using local exhaust ventilation to reduce the exposure [1], [2]. WorkSafe Western Australia also suggests that the extraction system should be placed within 100 mm of the soldering point so that the fumes can be removed properly [3]. However, in many normal laboratory and hobby soldering stations, proper automatic fume extraction is still not available and mostly the user has to control the ventilation by themselves.

Different research has also been done to study the fumes and particles produced during soldering. A study published in 2025 used Laser-Induced Breakdown Spectroscopy (LIBS) to analyze the amount of heavy metals and particles produced at different soldering temperatures [4]. The study found that when the soldering temperature increases, the amount of lead, tin, PM2.5 and other suspended particles also increases. This shows that soldering temperature has a direct relation with the amount of harmful fumes and particles produced. Because of this reason, temperature monitoring is also important in our project. We use the MAX6675 with a K-type thermocouple to monitor the soldering iron temperature instead of depending only on the gas sensor.

Another study in 2025 used single-particle aerosol mass spectrometry with a machine-learning method to analyze soldering smoke particles [5]. The system was able to classify the particles according to their size and chemical composition with an accuracy of 93.75%. For our project, the main important point from this study is that soldering fumes change depending on the soldering condition and temperature. This supports our use of the MQ-135 gas sensor for detecting changes in the air condition instead of depending on only one fixed sensor value.

Another research also studied tin-fog aerosol formation during fluxless soldering and showed that even soldering without normal flux can still produce metallic particles in the

air [6]. So, these studies show that proper fume monitoring and extraction is important during soldering.

B. Protocol Analysis

Different communication protocols are used in this project for connecting the sensors and other components with the STM32. I2C communication allows multiple devices to communicate using the same two communication lines [7]. In our project, the MPU-6050 accelerometer and SSD1306 OLED display are connected through the I2C bus. The MPU-6050 is used to measure the movement of the soldering iron, while the OLED display is used to show the current system condition. According to the MPU-6050 register information, sensor data can be read through I2C within a short amount of time, which makes it suitable for continuous monitoring [8].

SPI communication is used for the MAX6675 thermocouple module because the MAX6675 provides its temperature data through an SPI-compatible interface [9], [10]. It gives a 16-bit data output and provides temperature information with a resolution of 0.25 °C. In this project, the STM32 reads the MAX6675 data through SPI and calculates the temperature in degree Celsius by using the received raw value. The measured temperature is then compared with a safety threshold of 40 °C to understand whether the soldering iron is still hot or safe to touch.

The MQ-135 gas sensor provides an analog voltage depending on the air condition, so ADC is used to read the sensor value. The sensor reading can change because of temperature, humidity and also the condition of the sensor over time [11], [12]. Because of this, using only one fixed threshold may not always give proper result. To solve this problem, we use dynamic calibration in our project. When the system starts, the MQ-135 first gets 15 seconds for warm-up and after that multiple samples are taken to calculate the normal air baseline. The baseline is also updated slowly when the soldering iron is docked, so the system can adjust with changes in the environment. GPIO is also used for the FC-51 docking sensor and for controlling the L293D motor driver of the exhaust fan. PWM from TIM1 is used for controlling the SG90 servo motor [19].

C. Hardware Architecture Context

Different research works have shown the importance of using sensors and automatic control in soldering systems. A study in 2024 used pressure and temperature sensors with an IoT monitoring system in an industrial vapour-phase soldering oven [13]. The sensor information was used to control the heating process, and the system was able to reduce unnecessary heating time by around 60 seconds for each cycle. This also reduced the energy consumption by around 15–20%. The basic idea of that system is similar with our project because sensor data is taken first, then a controller processes the data and finally an actuator takes action. In our project, for example, the MAX6675 sends temperature information to the STM32 and based on different conditions the STM32 controls the fan and servo.

Another research in 2024 studied a packaging method for FBG temperature sensors by using ultrasonic-assisted soldering [14]. The study showed good mechanical strength of the sensor attachment. This shows that proper temperature sensing and sensor placement are also important for getting reliable temperature data during soldering.

Other studies also focused on controlling temperature during different soldering processes. Research on laser soldering of Al/Cu flexible PCB joints [15] and light-based reflow soldering [17] showed that proper heat control can improve the soldering process and reduce different problems in the solder joint. Another study used localized electromagnetic induction heating for very small soldering joints and completed soldering of around 170 μm solder balls in approximately three seconds by using controlled heating [18]. These studies show that temperature monitoring and proper control are important in soldering systems. However, a normal soldering iron mostly depends on the user and does not provide automatic monitoring of fumes, movement or docking condition. Our project adds this monitoring by using different sensors with the STM32.

D. Gaps Addressed by This Design

From the previous studies, some limitations can be found which our project tries to address. First, many studies discuss the health risk of soldering fumes and recommend proper ventilation [1]–[6], but most normal soldering stations still depend on the user to turn the ventilation system on or off. In our project, different conditions such as temperature, fumes, soldering iron movement and docking state are monitored together and the exhaust fan and ventilation system can be controlled automatically.

Second, many advanced soldering systems use sensors and actuators for monitoring different conditions [13]–[18], but usually they focus on only one or two parameters. In our project, several sensors and communication protocols are combined in one system. ADC, I2C, SPI, GPIO and PWM are used with the STM32F103C8T6 to collect data and control different parts of the system.

Another limitation is related to the MQ-135 gas sensor. The reading of this sensor can change because of environmental condition and sensor aging [11], [12]. Many simple systems use a fixed threshold for detecting gas or fumes, which may not work properly in every environment. For this reason, our project uses an automatic baseline calibration and continuously adjusts the baseline when the soldering iron is docked. This helps the system to detect changes in the air condition according to the normal environment instead of depending only on one fixed value.

In addition to these functions, our system also provides remote monitoring through a web-based dashboard. The sensor information can be sent through UART to the Wemos module and then published through MQTT. The website can show temperature, fume condition, docking status and movement condition in real time. It also gives warning to the user when the soldering iron remains undocked and without movement for a long period. Therefore, the project combines local safety

control and remote monitoring in one system, which makes the soldering workbench more safe and easier to monitor.

III. Design Methodology

The system was engineered using a modular co-design approach, partitioning real-time deterministic tasks from network bound cloud processes. Development progressed through three distinct phases:

- **Hardware & Circuit Prototyping:** Physical wiring and breadboard integration of environmental sensors, output indicators, motor drivers, and microcontrollers.
- **Local Firmware Implementation:** Development of low-level driver configurations, sensor sampling algorithms, threshold control logic, and peripheral signaling.
- **Network & Cloud Integration:** Implementation of the serial communication bridge, data parsing, cloud messaging protocols, and front-end dashboard hosting.

A. System Architecture

The platform utilizes a layered edge-to-cloud architecture. At the local edge layer, an ARM Cortex-M3 processor (STM32 Blue Pill) functions as the core control unit, executing real-time sensor reads and driving immediate hardware peripherals—including the OLED display, L293D fan motor driver, and active buzzer.

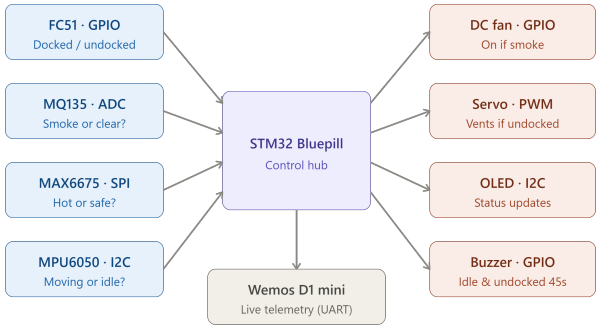


Fig. 1. Hardware Level Control Flow Diagram

To prevent network latency from blocking critical control loops, communication tasks are offloaded to a Wemos D1 Mini (ESP8266) gateway via serial linkage. The Wemos routes processed telemetry to a HiveMQ MQTT broker to feed a real-time web interface, while concurrently issuing asynchronous HTTP POST requests to a Google Apps Script web app for historical database logging in Google Sheets.

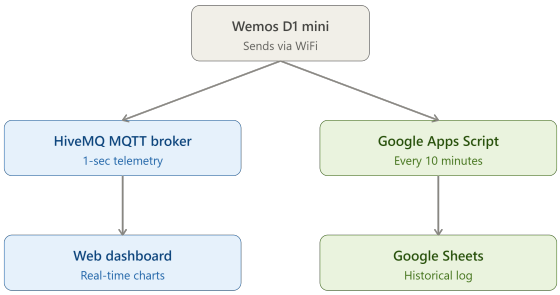


Fig. 2. Network Control Flow Diagram

B. Hardware Interfacing

The hardware integration relies on standard communication protocols and peripheral control interfaces to balance deterministic edge execution with remote telemetry transmission:

- **UART Protocol (USART2):** Established across pins PA2 (TX) and PA3 (RX). The STM32 transmits raw sensor data as a compact, comma-separated string over the serial bridge. The Wemos D1 Mini receives this stream, parses the delimiter, and restructures the values into a structured JSON payload for cloud transport.
- **I2C Protocol:** Configured on the STM32 over dedicated SDA and SCL lines to drive the OLED display module, enabling continuous, low-latency local visual telemetry.
- **SPI Protocol:** Implemented for high-speed synchronous serial data transfer between the STM32 and dedicated hardware modules (such as external storage interfaces or high-frequency SPI sensors), utilizing hardware SPI clock (SCK), MISO, and Chip Select (CS) lines.
- **GPIO and PWM Actuation:** Discrete digital GPIO pins trigger the active buzzer for acoustic threshold alarms, while PWM signals dynamically regulate the L293D motor driver for automated fan speed control.
- **Sensor Calibration & Sampling:** Raw physical signals are sampled by the STM32, passing through software-level noise filtering and offset calibration steps prior to local display rendering and UART serialization.

C. Tools and Resources

System development leveraged an Ubuntu Linux environment across embedded, network, and full-stack software domain tools.

Operating System: Ubuntu OS

Hardware Components:

TABLE II
Component List

Sensor	Protocol	Actuator	Protocol
FC-51	GPIO	OLED Display	I2C
MQ-135	ADC	SG90 Servo Motor	PWM
MAX6675	SPI	5V DC Fan	GPIO
MPU-6050	I2C	Active Buzzer	GPIO

Coding Environment:

Tool	Utility
STM32CubeMX	

TABLE III
SOFTWARE TOOLS

Cloud & Web Infrastructure:

- **Front-End:** Custom HTML/CSS/JavaScript dashboard hosted on GitHub Pages.
- **Backend & Storage:** HiveMQ MQTT Broker (real-time telemetry) and Google Apps Script mapped to Google Sheets (persistent database).

IV. DESIGN OUTCOME

V. PROJECT TIMELINE

The entire project was accomplished in 6 weeks, where 2 weeks were taken for the initial planning and the final testing and validation, with the other 4 taken for the development.

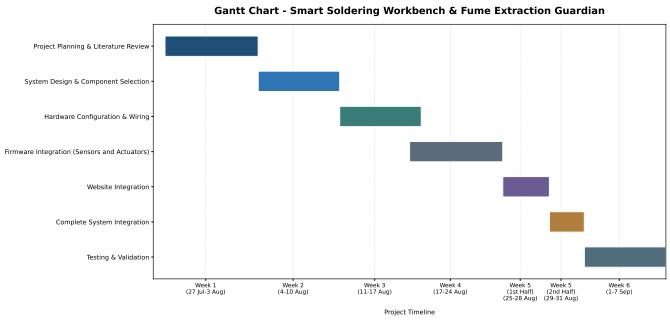


Fig. 3. Project Timeline Gantt Chart

VI. BUDGET

The project cost was distributed entirely into the hardware components, as the software components were open-sourced. For the firmware environment setup, open-source STM32CubeMX and STM32CubeIDE were utilized. For the website, GitHub Static Website was hosted in GitHub for free access, along with free-of-cost Google Sheets Database and HiveMQ MQTT Broker.

TABLE IV
COST ESTIMATION OF PROJECT COMPONENTS

SL No.	Component	Cost (BDT)
1	Breadboards	200
2	Wires	200
3	STM32 Bluepill	450
4	ST-LINK V2	400
5	FC-51	60
6	MAX6675	500
7	MQ-135	100
8	MPU6050	500
9	Servo Motor	100
10	DC Fan	100
11	Buzzer	100
12	OLED Display	250
	Total Cost	**2,960**

VII. TEAM MEMBERS AND ROLES

VIII. RISKS AND MITIGATION STRATEGIES

IX. CONCLUSION AND FUTURE WORK

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